

Investment Idea: Copper Supply De-Risking via Royalty Acquisition

Theme: Critical Minerals

Sector: Mining

Target: Real Assets with Commodity Price Upside

Investment Bottom Line: NSR royalties on producing copper mines deliver 10 - 15% net IRR with zero operating cost exposure. They capture full commodity price upside in a structural deficit where EV, renewable, and grid demand will outpace supply by 5–10 million tons annually by 2030–2035, scaling cash flows automatically and perpetually as prices rise.

Qualitative Analysis (*Catalyst, Driver, Tailwinds and Risks*)

Strategic Context

- **Supply Deficit:** Global copper grades declined from 1.5% to 0.5% since 1900, requiring 3x more ore processing. Goldman Sachs, Wood Mackenzie, and S&P Global project a 5–10 Mtpa structural deficit by 2030–2035.
- **Long Lead Times:** A 16-20 year discovery-to-production timeline means undiscovered copper cannot hit the market before 2040+.

Tailwinds

- **Electrification Demand:** The IEA projects copper demand to grow 2.5x by 2050. EVs require 80 - 100 kg of copper vs. 20-25 kg for internal combustion engines.
- **Policy Support:** Copper's U.S. Critical Minerals List designation accelerates domestic permitting and unlocks Defense Production Act funding.
- **Declining Supply:** Major production centers, such as Chile's Escondida and Collahuasi mines, face declining output from ore grade deterioration.

Catalyst & Driver

- **Catalyst:** The IEA confirms copper as the single most supply-constrained metal for the energy transition; demand is mathematically driven by the physics of electrification.
- **Driver:** A 3% NSR royalty receives top-line revenue before any operating cost, capex, or labor deductions - insulating returns even if mine operating costs spike 50%.

Risks to Making this Investment:

- **Mine Operator Risk:** Poor operator decisions or under-capitalization can reduce production volumes and directly lower royalty cash flows.
- **Resource Nationalism:** Core reserve regions (Chile, Peru, DRC) have a history of unilateral royalty increases, windfall taxes, and contract renegotiations.
- **Tailings Dam Failure:** Catastrophic collapses can cause permanent mine closure, eliminating royalty income (though physical liability remains entirely with the operator)

Qualitative Analysis Bottom Line: Copper royalties represent an elegant real asset structure, offering full participation in a highly certain commodity price appreciation thesis, driven by structural supply deficits and mandatory electrification demand while bearing zero operating cost, capex, or labor risk. The royalty holder secures a pure, perpetual income stream from a physically essential resource that cannot be substituted in the energy transition.

Quantitative Analysis (*Market Size, Growth & Return Expectations*)

Market Size and Growth

- **Structural Supply Deficit:** S&P Global projects a 9.9M ton annual copper deficit by 2035 - the equivalent of 40–50 non-existent world-class mines.
- **Royalty Transaction Market:** Exceeded **\$15B in 2024**; pension and sovereign wealth funds are increasingly bypassing intermediaries to acquire NSR royalties directly.
- **Copper Price Trajectory:** Consensus forecasts \$5.00–7.00/lb by 2030 (vs. current ~\$4.20). Each \$1/lb increase adds **~\$26M annually** to a 3% NSR on a 200kt/year mine.

Return Expectations

- **Target Net IRR:** 10 - 15% based on \$4.50–5.50/lb copper; mechanical upside with every \$1 price increase.
- **Revenue Math:** 3% NSR on a 200kt/year mine yields \$53M/year (at \$4.00/lb), \$66M/year (at \$5.00/lb), and \$79M/year (at \$6.00/lb) with zero operating cost exposure.
- **Duration:** 30 - 50+ years (life-of-mine); ideal perpetuity-equivalent for pension asset-liability management. Cash flows commence immediately.
- **Portfolio Construction:** 10 - 30 royalties across 3+ jurisdictions and multiple operators to mitigate single-mine risk while keeping full copper price exposure.

Quantitative Analysis Bottom Line: Copper royalties combine perpetuity-duration cash flows, full commodity price optionality, and zero operating cost exposure at core-plus risk. In a 2030 - 2035 structural supply deficit, a 3% NSR generates 50% more annual cash at \$6/lb copper than at \$4/lb with zero incremental cost, delivering pure price leverage with zero operational risk.